CST8918
DevOps - Infrastructure as Code

Course Outline

2025-2026

Pre-requisite(s)	N/A
Co-requisite(s)	N/A
Prepared by	John Dallas
Approved by	John Dallas, Chair
Approval date	Thursday, December 18, 2025
Normative hours	42.00
Grading system	A+ Through F

Applicable Program(s)	Level	Core/Elective
1531X01FWO - Cloud Development and Operations	2	Core
1531X03FWO - Cloud Development and Operations	2	Core
1531X07PAO - Cloud Development and Operations	2	Core
1531X09FAO - Cloud Development and Operations	2	Core

Land Acknowledgement

Algonquin College campuses in Ottawa, Perth and Pembroke are located on the traditional unceded, and unsurrendered territory of the Anishinàbe Algonquin People. The Algonquin People have inhabited and cared for these lands since time immemorial. We take this time to express our gratitude and respect to them and to the land for all that it has provided and will continue to provide.

As a post-secondary institution, we acknowledge the harms done to Indigenous Peoples and are committed to learning from the past. We pledge to promote healing and resilience as we move forward in partnership with the Algonquin Nations, First Nations, Métis, and Inuit Peoples in a spirit of reconciliation.

While we recognize that territorial acknowledgements are only one step in cultivating greater respect and inclusion of Indigenous Peoples, we commit to accompanying these words with actions. We are dedicated to building a future and community that is better for all.

We pledge to continue exploring and making meaningful contributions to the Truth and Reconciliation Commission of Canada's Calls to Action.

Learn more about the College's commitment to Truth and Reconciliation: https://www.algonquincollege.com/tri

Course Description

In the cloud computing model, all traditional infrastructure elements, such as servers, networks and firewalls, are virtualized. These components are defined in code and can then be deployed or decommissioned on-demand. Without systematic controls and policy enforcement mechanisms, this flexibility can result in inconsistent performance and wasted resources. Students employ industry-standard tools to manage both application code and the virtualized deployment environment. Industry best practice recommendations guide lab scenarios for automating the continuous integration and continuous deployment (CI/CD) of application code and automating infrastructure management tasks such as auto-scaling, container orchestration, and data extract-load-transform (ETL) tasks.

Vocational Learning Outcomes

This course provides the opportunity for you to achieve the following outcomes:

1531X01FWO - Cloud Development and Operations

VLO 3 Design and implement highly-available and elastically-scalable solutions within cloud computing platforms to maximize the reliability and resiliency of application services. (T, CP)

VLO 4 Integrate and automate development and operations processes to enhance business innovation and agility with consistently reliable solutions within an established policy framework. (T, CP)

VLO 5 Evaluate and integrate industry best practice recommendations into the design and development of secure cloud-based solutions to protect the privacy and integrity of client data. (T, CP)

VLO 6 Evaluate the full cost implications of various cloud solution options to find an optimal cost/benefit balance for an organization. (T, A)

VLO 7 Assess and implement policy monitoring and compliance strategies and tools to ensure all deployed cloud-based solutions adhere to relevant regulatory, governance and legal requirements. (T, A)

1531X03FWO - Cloud Development and Operations

VLO 3 Design and implement highly-available and elastically-scalable solutions within cloud computing platforms to maximize the reliability and resiliency of application services. (T, CP)

VLO 4 Integrate and automate development and operations processes to enhance business innovation and agility with consistently reliable solutions within an established policy framework. (T, CP)

VLO 5 Evaluate and integrate industry best practice recommendations into the design and development of secure cloud-based solutions to protect the privacy and integrity of client data. (T, CP)

VLO 6 Evaluate the full cost implications of various cloud solution options to find an optimal cost/benefit balance for an organization. (T, A)

VLO 7 Assess and implement policy monitoring and compliance strategies and tools to ensure all deployed cloud-based solutions adhere to relevant regulatory, governance and legal requirements. (T, A)

1531X07PAO - Cloud Development and Operations

VLO 3 Design and implement highly-available and elastically-scalable solutions within cloud computing platforms to maximize the reliability and resiliency of application services. (T, CP)

VLO 4 Integrate and automate development and operations processes to enhance business innovation and agility with consistently reliable solutions within an established policy framework. (T, CP)

VLO 5 Evaluate and integrate industry best practice recommendations into the design and development of secure cloud-based solutions to protect the privacy and integrity of client data. (T, CP)

VLO 6 Evaluate the full cost implications of various cloud solution options to find an optimal cost/benefit balance for an organization. (T, A)

VLO 7 Assess and implement policy monitoring and compliance strategies and tools to ensure all deployed cloud-based solutions adhere to relevant regulatory, governance and legal requirements. (T, A)

1531X09FAO - Cloud Development and Operations

VLO 3 Design and implement highly-available and elastically-scalable solutions within cloud computing platforms to maximize the reliability and resiliency of application services. (T, CP)

VLO 4 Integrate and automate development and operations processes to enhance business innovation and agility with consistently reliable solutions within an established policy framework. (T, CP)

VLO 5 Evaluate and integrate industry best practice recommendations into the design and development of secure cloud-based solutions to protect the privacy and integrity of client data. (T, CP)

VLO 6 Evaluate the full cost implications of various cloud solution options to find an optimal cost/benefit balance for an organization. (T, A)

VLO 7 Assess and implement policy monitoring and compliance strategies and tools to ensure all deployed cloud-based solutions adhere to relevant regulatory, governance and legal requirements. (T, A)

Assessment Levels —T: Taught **A**: Assessed **CP**: Culminating Performance

Course Learning Outcomes / Elements of Performance

When you have earned credit for this course, you will have demonstrated the ability to:

1. Prepare reusable infrastructure deployment scripts for environments appropriate to various application development lifecycle stages. (4)

1.1 Create infrastructure deployment scripts for different stages of the application development lifecycle.
1.2 Implement version control for infrastructure deployment scripts, using tools as Git.
1.3 Validate infrastructure deployment scripts using automated testing tools such as Terraform and Ansible

2. Compare various open-source tools for automating continuous integration and continuous deployment (CI/CD) with similar cloud vendor-provided tools. (4,6,7)

2.1 Identify key features and benefits of open-source CI/CD tools.
2.2 Comparing and contrasting the tools based on factors such as ease of use, customization options, community support and cost.
2.3 Evaluate cloud vendor-provided CI/CD tools.
2.4 Implement a CI/CD pipeline using open-source tools.
2.5 Integrate cloud vendor-provided CI/CD tools with infrastructure as code (IaC) framework.

3. Compose automation scripts that scale deployed application components in response to variable workload demand. (3,4)

3.1 Design scalable architecture for applications that can handle variable workload demands.
3.2 Develop automation scripts for scaling application components using tools as Terraform and Ansible to response to variable workload demand.
3.3 Implement continuous monitoring and optimization strategies for applications that are deployed using automation scripts.

4. Assess the efficiency of various data extract-load-transform (ETL) solutions for a given workload. (5,6,7)

4.1 Evaluate different ELT solutions based on workload requirements.
4.2 Implement ELT solutions using IaC tools such as Terraform and Ansible.
4.3 Monitor and optimize ELT workflows to ensure that they are efficient and effective.

Learning Resources

This course is part of a **B**ring **Y**our **O**wn **D**evice program, students are required to have access to a device that meets program requirements.

Recommended Resources:

- Morris Kief. (2020). *Infrastructure as Code*. O'Reilly Media, Inc.
- Brikman Yevgeniy. (2022) *Terraform: Up and Running*. O'Reilly Media, Inc.
- Meijer Bas, Hochstein Lorin, Moser René. (2022) *Ansible: Up and Running*. O'Reilly Media, Inc.
- Mehraban Yaser, Downs John, Rosini Edmondo. (2022) *Infrastructure as Code with Azure Bicep*. Packt Publishing.

To access your course list, learning resources and costs, visit Booklist.
Please note that a learner must be registered in a specific course or program to view this information.
Program tuition and fees can be found on the Tuition and Fees Estimator for Full-time Programs page.

Learning Activities

During this course, you are likely to experience the following activities:

- Classroom lectures
- Self-directed readings and research (online materials)
- Hybrid learning activities, including readings, discussion boards, pre-lab preparations.
- Class discussions
- In-lab exercises, individually or in groups.
- Quizzes
- Tests

Pre-defined Evaluation / Earning Credit

The following list provides evidence of this course's learning achievements and the outcomes they validate:

Lab Activities and/or Practical Assessments (30%)

Validates Outcomes: CLO 1, CLO 2, CLO 3, CLO 4

Quiz(zes) (10%)

Validates Outcomes: CLO 1, CLO 2, CLO 3, CLO 4

Midterm Exam(s) (15%)

Validates Outcomes: CLO 1, CLO 2, CLO 3

Final Practical Assessment (20%)

Validates Outcomes: CLO 1, CLO 2, CLO 3, CLO 4

Final Exam (15%)

Validates Outcomes: CLO 1, CLO 2, CLO 3, CLO 4

Oral Exam (10%)

Validates Outcomes: CLO 1, CLO 2, CLO 3, CLO 4

Prior Learning Assessment and Recognition

Students who wish to apply for Prior Learning Assessment and Recognition (PLAR) need to demonstrate competency at a post-secondary level in all outlined course learning outcomes. Evidence of learning achievement for PLAR candidates includes:

- Challenge Exam
- Performance Test

Course Related Information

Classroom Lectures:

Course material will be posted ahead of time; presentations, videos and chapter readings will be used to help students understand the required material.

Labs:
The labs are the hands-on component of the course, allowing students to apply relevant portions of the theoretical material in a more concrete way.

It is also important to follow the lab sessions closely as they do not necessarily “track” the theory lectures. The lab sessions are structured as independent stand-alone learning opportunities that complement the theory portion of the course.

Students are expected to perform initial analysis and design before their scheduled lab, in order to take advantage of the limited lab time. Laboratory assignments will be closely integrated with the theory material. The students’ ability to successfully complete the assigned exercises will directly correlate with their level of success on tests and the final exam.

This course requires you to learn material in the following broad areas listed below, and the organization of the theory and lab portions of the course will reflect this division of learning.

1. Concepts related to “how it works”, “why it's necessary”, “where/when it's used”.
2. Standards & practices related to applying the knowledge & technologies to achieve certain results.
3. Recognizing relevant and valid applications of the technology, and knowing what to
4. expect from them.
5. Ways and means of dealing with malfunctioning networks.
6. Evolutions and trends related to all the above.
7. The acronyms and technical jargon associated with all of the above.

The ICT Department requires that all course assignments (homework exercises, laboratory work, projects, etc) be submitted by students using a standard which could be specific to one or more courses. Professors will ensure, at the beginning of the term that students are advised of the exact details of these course specific submission requirements. Professors will also post them online alongside the course outline. Student submissions that do not meet the course published submission standards may not be marked, and may incur a penalty of up to 100% of the submission mark.

Grades:

The minimum passing grade for the course is 50% (D-). Course evaluation is based on a variety of evaluated submissions including projects, quizzes, labs, assignments, oral exam, midterm and final exams; however, to pass the course, students must also achieve a minimum grade of 50% on the Oral Exam, regardless of their overall course average.

Program Information

1531X01FWO - Cloud Development and Operations

Cloud Usage
Students in this program may be exposed to Cloud Software through student accounts but they should never use their Credit Card for these services. Algonquin College will not be responsible for a students' utilization of Online Services or Cloud Based Applications.

Appropriate Use of Generative Artificial Intelligence (GenAI) Tools and AI-Assisted Resources
In alignment with Algonquin College Policy AA48 (Academic Integrity), students must follow the specific evaluation instructions for each assessment and consult the professor if they are unsure whether GenAI tools are permitted. For the purposes of this course, "GenAI/AI-assisted resources" includes any tool, device, software, browser feature, plug-in, or service that can generate, rewrite, summarize, translate, solve, or provide suggested answers or content (e.g., chatbots, writing/code assistants, AI search tools, and AI-enabled devices such as smart glasses or similar wearable technology). Unless an assessment explicitly allows such tools/resources, students should assume they are **not permitted**. Where use is allowed, students must clearly acknowledge and/or cite that use as directed by the instructor. Submitting AI-generated or AI-rewritten content as one's own work, using AI outside the permitted scope, or failing to acknowledge/cite permitted use may be treated as academic misconduct under AA48.

1531X03FWO - Cloud Development and Operations

Cloud Usage

Students in this program may be exposed to Cloud Software through student accounts but they should never use their Credit Card for these services. Algonquin College will not be responsible for a students' utilization of Online Services or Cloud Based Applications.

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Department Information

STUDENT ACADEMIC RESPONSIBILITIES

Each student is responsible for:

- Knowing the due dates for marked out-of-class assignments.
- Attending all classes and knowing the dates of in-class marked assignments and exercises.
- Arriving to class on time and remaining for the duration of the scheduled class.
- Maintaining a folder of all work done in the course during the semester for validation claims in cases of disagreement with faculty.
- Keeping both paper and electronic copies of all assignments, marked and unmarked, in case papers are lost or go missing.
- Regularly checking both Brightspace announcements as well as one's Algonquin e-mail account for important messages from both professors and college administration.
- Participating in on-line and classroom exercises and activities as required.
- Retaining course outlines for possible future use to support applications for transfer of credit to other educational institutions.

Harassment/Discrimination/Violence will not be tolerated. Any form of harassment (sexual, racial, gender or disability-related), discrimination (direct or indirect), or violence, whether involving a professor and a student or amongst students, will not be tolerated on the college premises. Action taken will start with a formal warning and proceed to the full disciplinary actions as outlined in Algonquin College Policies - HR22 and SA07.

Harassment means one or a series of vexatious comment(s) (whether done verbally or through electronic means), or conduct related to one or more of the prohibited grounds that is known or ought reasonably to be known to be unwelcome/unwanted, offensive, intimidating, derogatory or hostile. This may include, but is not limited to: gestures, remarks, jokes, taunting, innuendo, display of offensive materials, offensive graffiti, threats, verbal or physical assault, stalking, slurs, shunning or exclusion related to the prohibited grounds.

For further information, a copy of the official policy statement can be obtained from the Student Association.

Violation of the Copyright Act
General – The Copyright Act makes it an offence to reproduce or distribute, in whatever format, any part of a publication without the prior written permission of the publisher. For complete details, see the Government of Canada website at http://laws.justice.gc.ca/en/C-42 . Make sure you give it due consideration, before deciding not to purchase a textbook or material required for your course.

Software Piracy - The Copyright Act has been updated to include software products. Be sure to carefully read the licensing agreement of any product you purchase or download, and understand the terms and conditions covering its use, installation and distribution (where applicable). Any infringement of licensing agreement makes you liable under the law.

Disruptive Behaviour is any conduct, or threatened conduct, that is disruptive to the learning process or that interferes with the well being of other members of the College community. It will not be tolerated. Members of the College community, both students and staff, have the right to learn and work in a secure and productive environment. The College will make every effort to protect that right. Incidents of disruptive behaviour must be reported in writing to the departmental Chair as quickly as possible. The Chair will hold a hearing to review available information and determine any sanctions that will be imposed. Disciplinary hearings can result in penalties ranging from a written warning to expulsion.

For further details, consult the Algonquin College Policies AA32, SA07 and IT01 in your Instaguide.

College Related Information

Algonquin College's policies have been developed to ensure the health, safety and security of all students, faculty and staff, and the proper and fair operation of the College as an academic institution and employer. Please refer to the Algonquin College Policies website for the most current policy information available at http://www.algonquincollege.com/policies/.

Students are especially encouraged to be aware of the following College expectations

Academic Integrity
Algonquin College is committed to the highest standards of academic integrity, and students are expected to uphold these standards throughout their learning journey with the College and post-graduation. Any academic work submitted by a student is expected to be their own work, unless designated otherwise, all human and technological assisted generated sources must be attributed. Refer to the College's Library web site for attribution support and resources. All students are expected to be familiar with the College's AA48: Academic Integrity policy which outlines student and staff roles and responsibilities, the process for addressing, determining and ruling final academic decisions regarding violations of academic integrity as well as what represents academic misconduct. Students with any questions about academic integrity course expectations including but not limited to use of technology, generative AI and other resources to support completion of coursework, are encouraged to speak to their professor and refer to the course weekly schedule, assessment instructions, course announcements and materials.

Centre for Accessible Learning (CAL)
Students with visible and/or non-visible disabilities are encouraged to register with the Centre for Accessible Learning (CAL) in order to be eligible for appropriate learning supports and/or accommodations. Students are strongly encouraged to make an appointment with the Centre for Accessible Learning as early as possible when starting a program. Once your needs are identified, a Letter of Accommodation (LOA) will be issued which you can share with your professors through the CAL Student Portal. If you are a returning student, please ensure that professors are given a copy of your LOA each semester.

Test Accommodations for Students Registered with CAL
Students who require testing accommodations will book their tests/exams through the CAL Student Portal. It is important to note the posted test/exam booking deadlines for the term. Test booking deadlines are communicated through CAL Disabilities Counsellors, CAL Testing Services, and other college wide communication. Additional information on the steps to book accommodated tests/exams with CAL Test Services can be found at Step-by-Step Overview.

Retroactive Accommodations
Students are expected to meet evaluation and completion deadlines as stated in course outline and weekly schedule documents. In circumstances where evaluation deadlines are missed or student performance has been affected by a temporary, significant and/or unanticipated change in functional ability related to disability (including mental health), interim or retroactive accommodations may be

considered. For these instances it is advised you reach out to CAL or your professors as soon as possible. For other situations where deferral of evaluations may be warranted, please consult Algonquin College Policy AA21: Deferred Evaluation.

Network Credentials and College Email

Students at Algonquin College are provided with a college email account. This is the address that will be used when the College, your professors, or your fellow students communicate important information about your program or course activities. Your network credentials can be found in the ACSIS portal and you are expected to check your Algonquin email regularly and to use it to send and receive college-related email. Students are required to maintain the privacy of their AC login credentials. Sharing network credentials poses a security risk and is subject to disciplinary action. Your responsibility to protect your login credentials is governed by the Information Security Policy IT01. Support is available through the college Information Technology Service (ITS) at: https://www.algonquincollege.com/its/

Mental Health and Wellness Supports

Canada has launched a national Suicide Helpline at www.988.ca and a wide range of mental health and wellness resources for Algonquin students are available at www.algonquincollege.com/SSS/student-health-wellness.

Student Course Feedback

Algonquin College invites students to share their course experience by completing a student course feedback survey for each course they take. For further details consult Algonquin College Policy AA25: Student Course Feedback.

Use of Mobile Devices in Class

With the proliferation of small, personal mobile devices used for communications and data storage, Algonquin College believes there is a need to address their use during classes and examinations. During classes, the use of such devices unless authorized by your professor can be disruptive and disrespectful to others. During examinations, the use of such devices is generally prohibited unless authorized by your professor. Otherwise use is considered academic dishonesty in the form of cheating. For further details consult Algonquin College Policy AA32: Use of Electronic Devices in Class

Technology Requirements

Students are required to have access to a computer and to the internet. There may also be additional technology-related resources required to participate in a course that are not included in the course materials fee, such as headphones, webcams, specialized software, etc. Details on these requirements can be found in the Weekly Schedule document of the course for each course available on Brightspace.

Transfer of Credit

It is the student's responsibility to retain course outlines for possible future use to support applications for transfer of credit to other educational institutions.

Safe Harbour

In the event of an unexpected major event (pandemic, etc.), your course may have changes that are not reflected in the Course Outline. Should this happen, the Weekly Schedule document will have updated information about your course.